

EE595H: Quiz

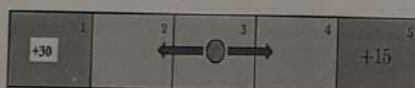
Total Questions: 2, Total Marks: 20, Time: 60 Mins

1. Let $\{X_n : n = 0, 1, \dots\}$ be a Markov chain with state space $\{0, 1, 2\}$ and transition probability matrix

$$P = \begin{bmatrix} 1/2 & 1/4 & 1/4 \\ 2/3 & 1/3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Starting from 0, what is the probability that the process never enters 1?

2. Consider a simplified game problem defined over the grid shown below:



The figure shows two terminal states $s \in \{1, 5\}$ with rewards $r(1) = +30, r(5) = +15$, and three additional states $s \in \{2, 3, 4\}$ with constant reward $r(s) = -1$. The terminal squares lead to the EXIT state where the game stops. In the center states, the agent can select equally between two actions: $a \in \{\text{RIGHT}, \text{LEFT}\}$, and the motion occurs according to this action without any randomness. Consider a generic discount factor $\gamma \in [0, 1]$. For what value(s) of γ , does the optimal action at state 4 become "moving left"?

10

EE595H (Stochastic Models): Half Semester Examination

Total Questions: 2, Total Marks: 25, Time: 120 Mins

1. Alex is an avid sports fan who watches football games regularly. His friend has observed that whether Alex watches a football game on a given day depends on the previous two games he watched, with the following probabilities:

- 7/8 if the last two games he watched were both football games,
- 1/2 if exactly one of the last two games was a football game,
- 1/8 if neither of the last two games was a football game.

- (a) Alex watched four games last week. If the first two were football, what is the probability that the fourth one was also a football game?

5

- (b) What is the long-run probability that Alex will watch two football games in a row?

5

Total: 10

2. Consider an MDP with three states $S = \{s_1, s_2, s_3\}$ and two actions $A = \{a_1, a_2\}$. The discount factor is given as $\gamma = 0.9$. The transition probabilities and rewards are as follows:

State	Action	Next State	Transition Probability	Reward
s_1	a_1	s_1	0.3	2
s_1	a_1	s_2	0.6	4
s_1	a_1	s_3	0.1	1
s_1	a_2	s_2	0.7	5
s_1	a_2	s_3	0.3	3
s_2	a_1	s_1	0.5	1
s_2	a_1	s_3	0.5	2
s_2	a_2	s_3	1.0	7
s_3	a_1	s_1	0.2	-1
s_3	a_1	s_2	0.8	0
s_3	a_2	s_3	1.0	0

Table 1: Transition Probabilities and Rewards

Initialize $v_0(s) = 0$ for all states. Using Bellman's optimality equation, compute the values $v_1(s)$ and $v_2(s)$. Also find the optimal deterministic policy after 2 iterations.

[Do the computations only up to 2 iterations]

15